AS · Edexcel · Physics

2 hours 11 questions

Structured Questions

Working as a Physicist

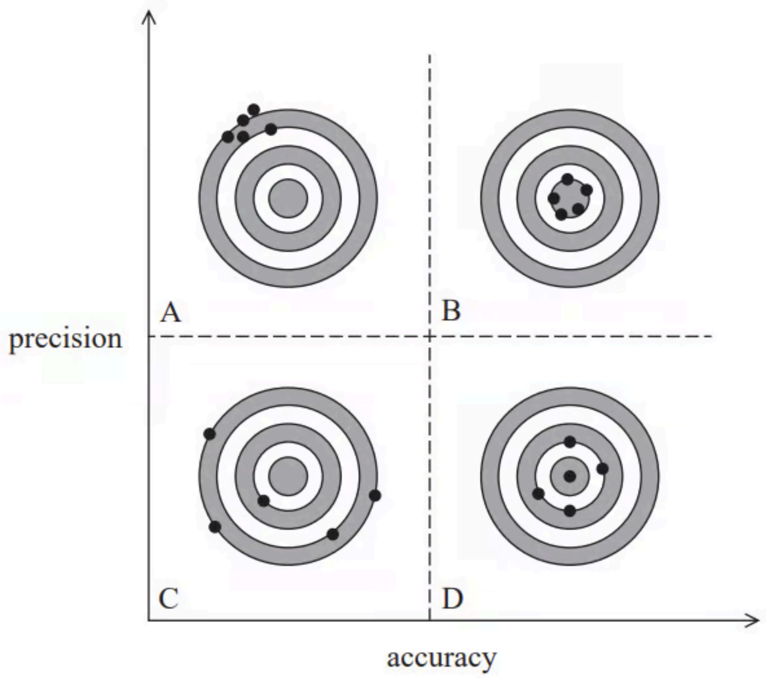
SI Units / Practical Skills / Estimating Physical Quantities / Limitations of Measurements / Scientific Communication / Applications of Science / The Scientific Community / Science & Society

Easy (2 questions)	/8
Medium (6 questions)	/64
Hard (3 questions)	/39
Total Marks	**/111**

Scan here to return to the course
or visit savemyexams.com

Easy Questions

1 A teacher is explaining the differences between accuracy and precision to her students. She draws the following diagram, which shows different degrees of accuracy and precision. The circles represent targets A, B, C and D and the dots represent arrows hitting the targets.



Explain how targets A, B, C and D represent differing degrees of accuracy and precision.

....

....

....

....

(4 marks)

2 (a) A practical physics textbook states that “measurements may give a precise value for the quantity being determined but this may not necessarily be an accurate value”.

Describe what physicists mean by the terms accuracy and precision.

(2 marks)

(b) The temperature of the air in a room is measured using a mercury-in-glass thermometer.

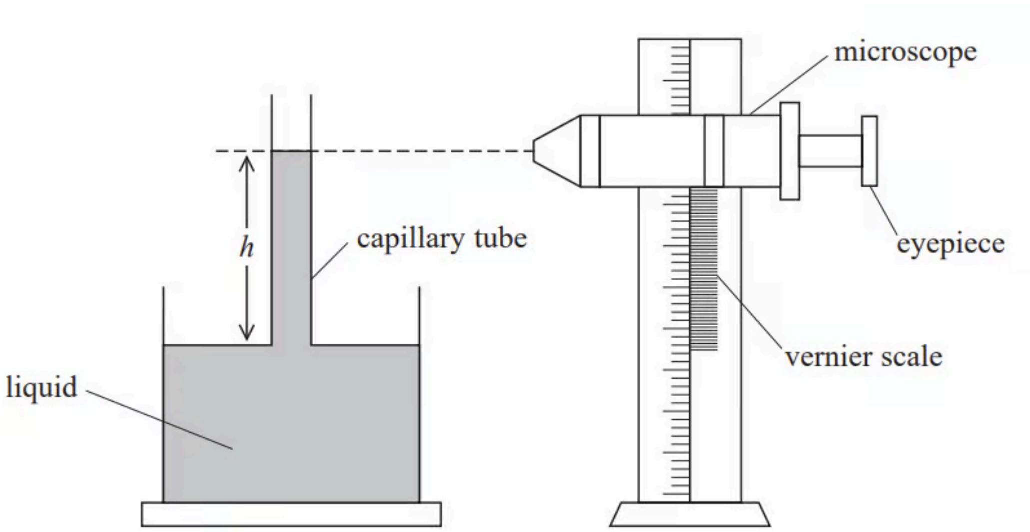
Describe how the value for the temperature may be precise but not accurate.

(2 marks)

Medium Questions

1 (a) A student measured the height h of a liquid column in a capillary tube. She used a travelling microscope to make measurements of the positions of the top and bottom of the liquid column.

The travelling microscope consists of a simple microscope that can be moved vertically along a vernier scale.



The student used a capillary tube with an internal radius *r* equal to 0.10 mm and recorded the following readings from the vernier scale.

Bottom of liquid column / cm	Top of liquid column / cm
12.00	27.10

i) State the uncertainty in each of these readings.

(1)

ii) Calculate the percentage uncertainty in the student's value of *h*.

Percentage uncertainty in h = ...**(2)**

iii) The student repeated the measurement of h for capillary tubes of different radii.

The table shows the student's final data.

***r* / mm**	**1 / r**	***h* / cm**
0.09	11.1	16.56
0.10	10.0	15.1
0.12	8.3	12.6
0.15	6.7	10.33

Criticise the student's recording of the data.

(2)

....

....

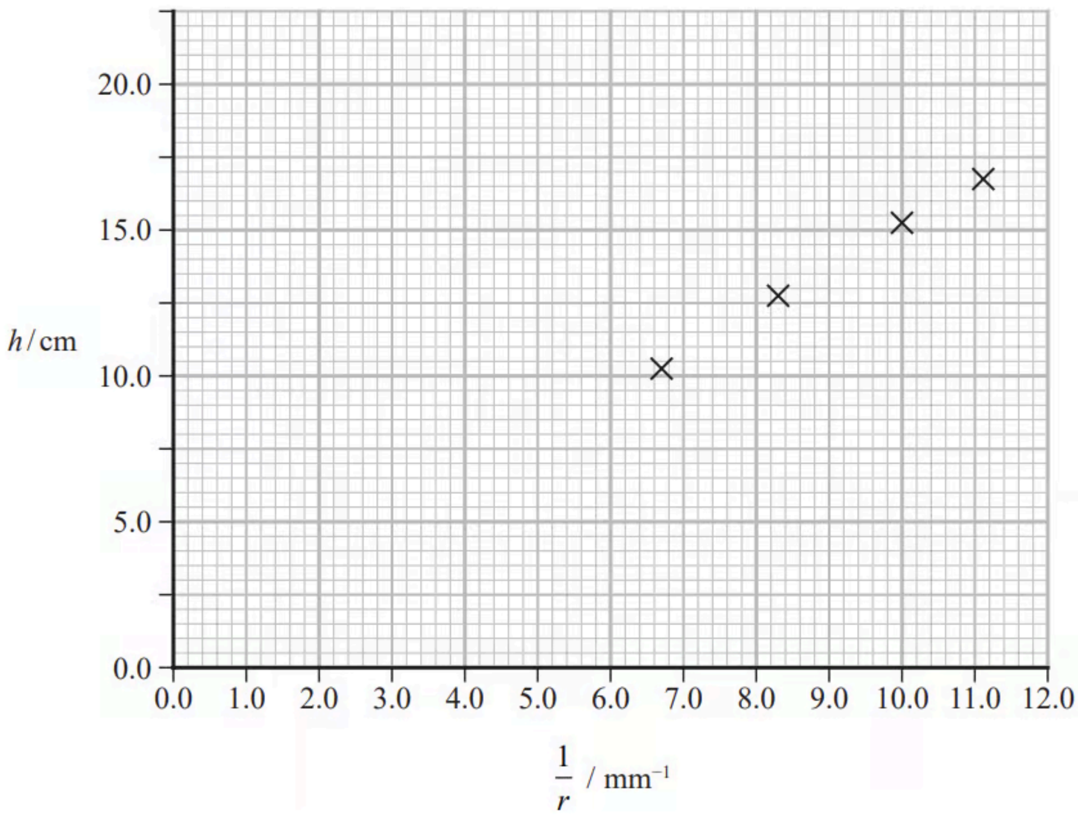
....

....

....

(5 marks)

(b) The student plotted the following graph.



i) Determine the height of the liquid column that the student could expect for a tube with an internal radius of 0.11 mm.

Height of liquid column = ... **(3)**

ii) In her notes it stated that

$$h = \frac{k}{r} \text{ where k is constant}$$

Assess the extent to which the student's data supports this relationship.

(4)

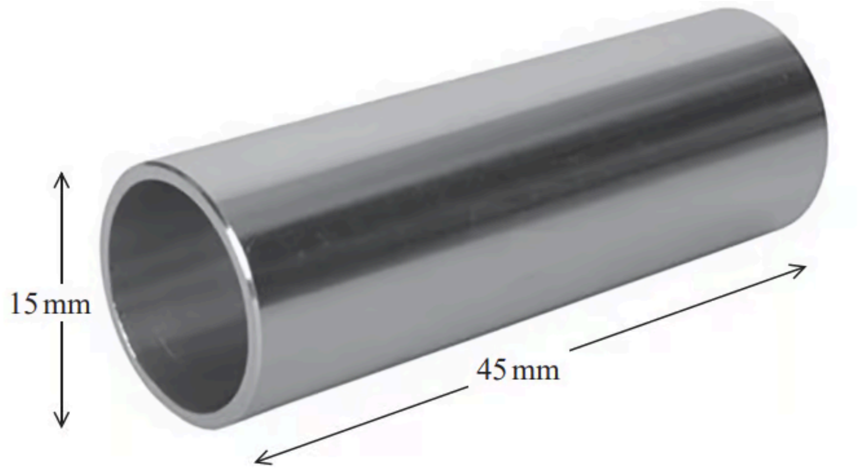
....

....

....

(7 marks)

2 (a) An engineer was checking the dimensions of a steel tube. The tube had a length of about 45 mm and an external diameter of about 15 mm as shown.



She used a digital micrometer to measure the diameter of the tube. Before taking the reading she closed the jaws of the micrometer to check for a zero error.

State the type of error she avoided by doing this.

....

(1 mark)

(b) Describe the procedure she should follow to determine an accurate value for the external diameter of the tube.

....

....

....

(3 marks)

(c) The engineer determined the length of the tube using the micrometer. The reading on the micrometer scale was 45.043 mm. She recorded the reading as 45.0 mm.

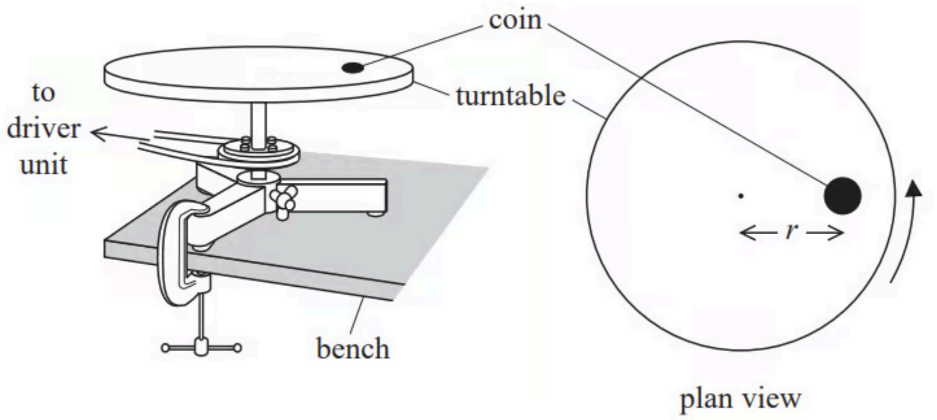
State why recording a reading of 45.043 mm could not be justified.

....

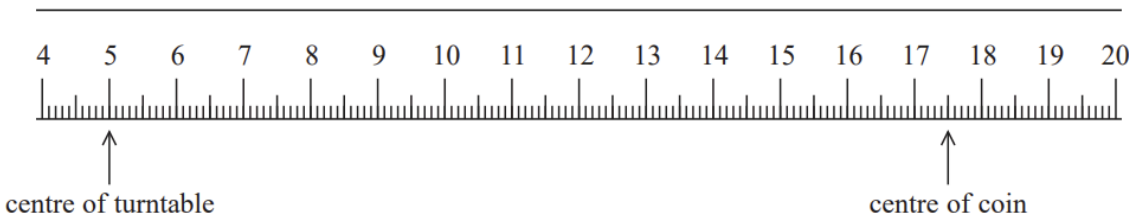
(1 mark)

3 (a) A student was investigating the forces involved in circular motion.

He placed a small coin on a horizontal turntable as shown. The turntable was connected to a driver unit so that it could be rotated at a constant rate.



The student measured the distance r between the centre of the turntable and the centre of the coin, with a metre rule as shown.



Explain why the percentage uncertainty in the value of *r* is about 1%.

Your answer should include a calculation.

....

....

....

(3 marks)

(b) The student switched on the driver unit and increased the rate of rotation until the coin slid off the turntable.

He read the angular velocity ω of the turntable from a digital display on the driver unit. He then replaced the coin in the original position on the turntable and repeated the procedure.

His results are shown.

ω / rad s^{-1}				
0.125	0.112	0.118	0.123	0.116

i) The student used the results to calculate a mean value of ω.

State the purpose of calculating a mean.

(1)

ii) Calculate the percentage uncertainty in the mean value of ω.

Percentage uncertainty = ...**(3)**

iii) The student used ω and r to calculate the centripetal acceleration of the coin at the instant it started to slide.

Calculate the percentage uncertainty in this centripetal acceleration.

Percentage uncertainty = ...**(3)**

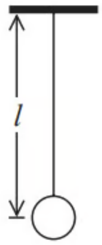
(7 marks)

(c) The student repeated the procedure with different values of r.

Explain how the value of ω at which the coin started to slide varied as r increased.

(3 marks)

4 (a) A student is using a simple pendulum to determine a value for the acceleration of free fall g.



She measures the length l of the pendulum four times with a metre rule and records the following values.

l / cm			
l_1	l_2	l_3	l_4
85.5	86.0	87.5	85.5

She calculates the mean length l_m of the pendulum using the following method:

$$l_m = \frac{85.5 + 86.0 + 87.5 + 85.5}{4} = 86.1 \text{ cm}$$

i) Calculate a more accurate value for l_m.

l_m = ...**(2)**

ii) Determine the time period of the oscillations of this pendulum, using your calculated value for l_m.

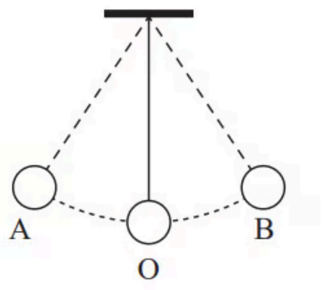
Time period of oscillations = ...**(2)**

....

....

(4 marks)

(b) She sets the pendulum into oscillations with small amplitude and uses a stopwatch to determine the time period.



The student releases the pendulum at A and simultaneously starts the stopwatch. She measures the time taken for 5 oscillations and divides the value by 5. She repeats the procedure twice and calculates a mean time period.

Explain two modifications to the student's method that would improve the value obtained for the time period.

(4 marks)

5 (a) The Beaufort scale is used to describe wind intensity. On this scale the average wind speed v increases with the Beaufort scale value *B*.

The relationship between v and *B* is given by

$$v = kB^p$$

where *k* and *p* are constants.

Explain why a graph of log v against log B should give a straight line.

....

....

(2 marks)

(b) The table gives some values of v and corresponding values of *B*.

v / m s^{-1}	*B*		
2.00	1		
10.0	3		
21.5	5		
36.0	7		
50.5	9		
68.0	11		

i) Plot a graph of log v against log *B* on the grid opposite.

Use the columns provided to show any processed data.

(5)

ii)

Determine the values of *p* and *k*.

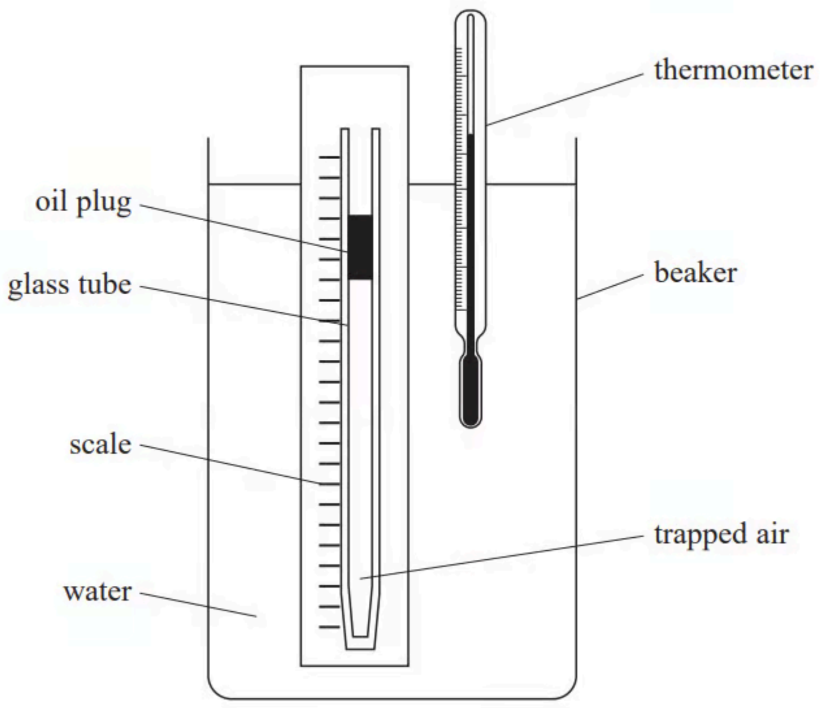
p = ...

k = ...

(3)

(8 marks)

6 (a) A student investigated how the volume of a fixed mass of air varies with the temperature of the air. She used the apparatus shown.



A glass tube was sealed at one end. A plug of oil trapped a length l of air in the tube.

The water in the beaker was heated to a temperature θ. The corresponding value of l was measured. This was repeated for a range of temperatures.

The thermometer had a resolution of 0.5 °C. The scale had mm divisions.

The student's results are shown in the table.

θ / °C	l / cm
24	8.8
60	9.8
78.5	10.3
95.5	10.9

i) Criticise the student's results.

(3)

ii) Explain two possible sources of error in this investigation.

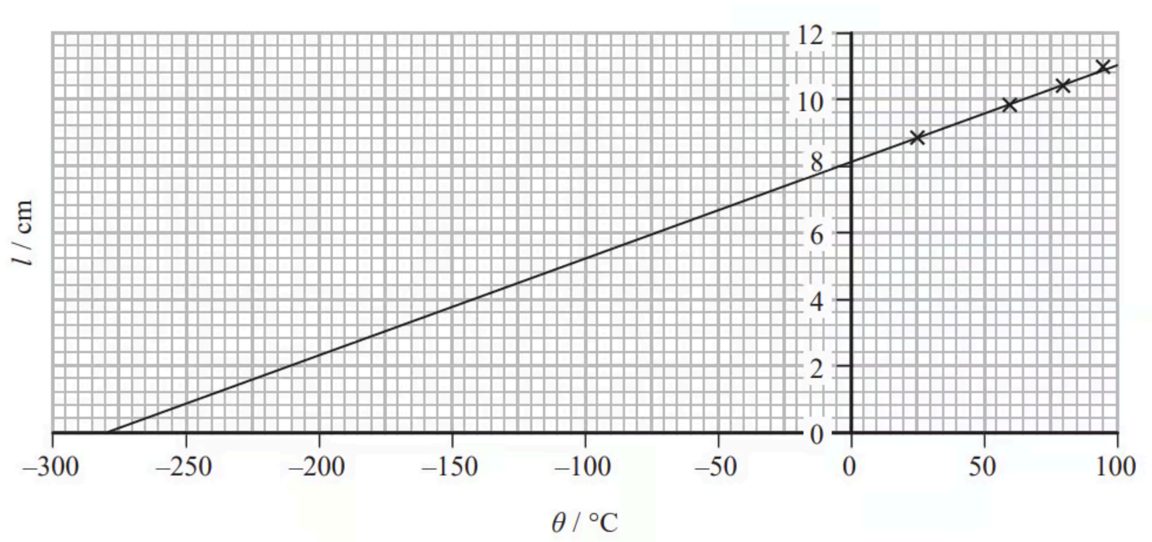
(4)

iii) Describe two improvements that would increase the accuracy of measurements obtained in this investigation.

(2)

(9 marks)

(b) The student plotted a graph of l against θ as shown.



i) Explain the significance of the intercept on the x-axis.

(3)

ii) The student wrote a report of the investigation in her lab book. In the conclusion she wrote:

"In this investigation uncertainties were minimised by selecting measuring instruments with a high resolution. The points lie on a perfect straight line, indicating that the investigation is accurate."

Discuss the student's conclusion.

(4)

(7 marks)

Hard Questions

1 (a) A spring is made from loops of thick steel wire as shown.

There are two extra loops, one on each end of the spring.

A student determined the length of steel used to make the spring by using vernier calipers to measure the width *w* of the spring.

The length of wire *l* on each loop is given by $l = \pi w$

The student obtained the following values for *w*.

***w* / mm**	15.3	15.2	15.4	15.3

i) Calculate *l*.

l = ...**(3)**

ii) Estimate the percentage uncertainty in your value for *l*.

% uncertainty in *l* = ...**(2)**

iii) Calculate the total length *L* of wire used to make the spring.

L = ...**(2)**

....

....

....

....

....

(7 marks)

(b) The student measured the diameter d of the steel wire and obtained a value of 2.52 mm.

i) Explain which instrument he used to measure the diameter.

(2)

ii) Estimate the percentage uncertainty in the student's value for *d*.

% uncertainty in *d* = ...**(1)**

iii) The student used a balance to measure the mass *m* of the spring.

He obtained a value of 32.0 ± 0.5 g.

Estimate the percentage uncertainty in the mass of the spring.

% uncertainty in *m* = ...**(1)**

iv) The student calculated the density ρ of the steel using the equation

$$\rho = \frac{m}{V}$$

Calculate the percentage uncertainty in his value for the density of steel.

% uncertainty in value for density of steel = ...**(1)**

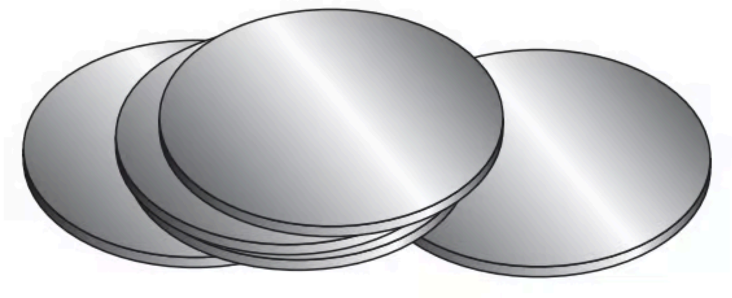
v) Determine whether the data collected leads to a value for the density of steel in agreement with the standard value.

density of steel = 7 800 kg m^{-3}

(4)

(9 marks)

2 (a) A student is investigating the properties of steel. He has fifty steel discs available.



Each disc has a diameter $d \cong 1.3$ cm and a thickness $t \cong 2$ mm.

State a suitable measuring instrument that could be used with a single disc to measure t.

....

(1 mark)

(b) A balance which can measure mass with a resolution of 0.2 g is available.

Determine the minimum number of discs that should be placed on the balance together if the percentage uncertainty in the measurement of the mass is to be less than 0.5%.

density of steel = 7900 kg m^{-3}

Minimum number of discs = ...

....

....

....

....

(4 marks)

(c) The measured uncertainty in d is ±0.1 mm and the measured uncertainty for t is ±0.05 mm.

Determine the percentage uncertainty in the calculated volume of the disc.

Percentage uncertainty in volume = ...

....

....

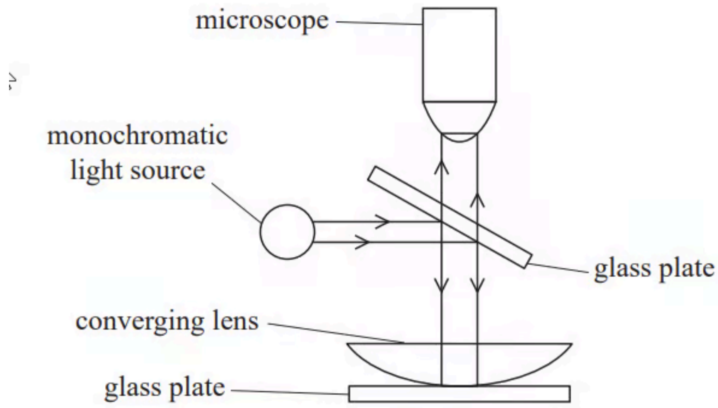
....

(3 marks)

3 (a) A method to determine the wavelength of light using a converging lens was first proposed by Sir Isaac Newton.

A converging lens is placed on a plane glass plate. The lens is illuminated from above with a parallel beam of monochromatic light, as shown.

Some of the light is reflected from the upper surface of the lower glass plate and some from the lower surface of the lens. Interference between these two reflected waves produces circular fringes. The pattern is viewed through a microscope.



Side view of apparatus

Pattern seen through microscope

The diameter *D* of each circular fringe, numbered *N* from the centre, is measured using the microscope. The data obtained from such an experiment is shown.

N	***D* / mm**		
1	5.13		
2	7.08		
3	8.71		
4	10.23		
5	11.48		

The relationship between N and D is of the form $D = pN^q$ where p and q are constants.

Determine p and q for this data using a graphical method. Use the additional columns for your processed data.

p = ...

$q =$ ....

....

....

....

....

....

(8 marks)

(b) The table below shows the readings from which the diameter of the first dark circle was calculated.

Position of left-hand side of circle / mm	Position of right-handside of circle / mm	Diameter / mm
54.79	49.66	5.13

i) Use these readings to estimate the percentage uncertainty in the diameter due to the resolution of the instrument.

Percentage uncertainty = ...**(2)**

ii) State why the actual percentage uncertainty would have been greater than the value calculated in (b)(i).

(1)

....

....

....

(3 marks)

(c) When considering the principles of this experiment, a student suggests that interference fringes would only be produced with monochromatic light. This is because interference requires coherent light waves.

Discuss the validity of the student's suggestion.

....

....

....

....

(4 marks)